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Evaluating the Waterfall Model and Designing the Black Adaptive Model

The Waterfall model is the oldest paradigm in software engineering. Pressman and Maxim (2020) describe it as a systematic, sequential approach that begins with customer requirements and moves through planning, modeling, construction, and deployment, ending with ongoing support of the completed software. It still fits projects with well-defined, stable requirements, but most modern projects do not look like that. This paper evaluates three limitations of the Waterfall model and then proposes the Black Adaptive Model, which keeps the five phases but runs them in short iterations connected by feedback loops.

Limitations of the Waterfall Model

The first limitation is frozen requirements. Waterfall assumes the customer can state every requirement at the start and that those requirements will stay stable. Pressman and Maxim (2020) point out that it is often difficult for customers to state all requirements explicitly at the beginning of a project, and that today's software work faces a constant stream of changes to features, functions, and content. A team that locks its requirements early ends up building to an outdated specification.

The second limitation is late testing. Testing happens in construction, after analysis and design are complete. Pressman and Maxim (2020) warn that major blunders may not be

detected until the working program is reviewed. A flawed design decision can sit unnoticed until code is built on top of it, so the fix reaches back across multiple phases when it costs the most.

The third limitation is late customer visibility. The customer approves a requirements document early but does not see working software until late in the project, and Pressman and Maxim (2020) note that the customer must be patient because of this. Customers often cannot tell whether a design meets their needs until they use it, so feedback arrives exactly when changes are hardest to make.

The Black Adaptive Model

The Black Adaptive Model keeps Waterfall's five framework activities but delivers the product in two-week iterations. Each iteration runs all five phases for a small increment and ends with a customer demo whose feedback starts the next iteration. This borrows the idea behind the spiral model, which Pressman and Maxim (2020) describe as coupling the iterative nature of prototyping with the controlled and systematic aspects of Waterfall. Hachemi (2020) calls this process pattern reuse for modification: Waterfall is the base model, and proven patterns from Agile, DevOps, and prototyping are merged in to fix its weaknesses.

In communication, the team meets with stakeholders at the start of every iteration to set that iteration's requirements and review the last demo's feedback. This fixes frozen requirements because requirements are revisited every two weeks. It also puts into practice a task Pressman and Maxim (2020) list for larger projects: packaging requirements so they can be delivered incrementally. Planning then breaks that work into tasks tracked on a DevOps board.

In modeling, the team designs the product flow and UML models for the current increment only. When a feature is hard to explain with a diagram, the team builds a prototype and the customer reviews it before construction starts. Pressman and Maxim (2020) explain that prototyping is more commonly used as a technique inside another process model than as a stand-alone model. The prototype is optional by design because prototypes cost time and can tempt developers into quick implementation compromises (Pressman & Maxim, 2020).

In construction, the team builds with continuous testing borrowed from DevOps, so tests run as code is written instead of at the end. A failing test leads to one of two loops. A code defect loops back to construction. A design flaw loops back to modeling. Separating them keeps a small bug from restarting the whole iteration.

In deployment, the team releases only when all tests pass, publishes release notes, and demos the working software to the customer. That feedback feeds the next communication phase, so the customer sees working software every two weeks instead of once at the end. Each of the model's four feedback loops traces back to a limitation, and the attached UML activity diagram shows each one as a labeled decision node.

Pressman and Maxim (2020) caution that a weak process hurts the product, but that an obsessive reliance on process is also dangerous. The Black Adaptive Model keeps the structure that makes Waterfall easy to plan and adds only the loops that solve a real Waterfall problem.

References

Hachemi, A. (2020). Software development process modeling with patterns. In *Proceedings of the 2020 2nd World Symposium on Software Engineering* (pp. 37–41). Association for Computing Machinery. https://doi.org/10.1145/3425329.3425339

Pressman, R. S., & Maxim, B. R. (2020). *Software engineering: A practitioner's approach* (9th ed.). McGraw-Hill Education.